

Three-step reduction to a finite SDP certificate

Continuous problem $\rightarrow$ simplex polynomial $\rightarrow$ Lasserre SDP $\rightarrow$ dual certificate

Continuous

$$C_{1a} = \inf_f \|f^* f\|_\infty / (\int f)^2$$

infinite-dimensional

Lemma 2.1

Simplex polynomial

$$\text{val}(d) = \min_{\mu \in \Delta_d} \max_W \mu^\top M_W \mu$$

polynomial minimization

Theorem 3.4

Lasserre SDP

$$\text{val}^{(k)}(d) \text{ via moment matrix } M_k(y) \succeq 0$$

finite SDP, dual feasible point

$$\text{soundness: } \text{val}^{(k)}(d) \leq \text{val}(d) \leq C_{1a}$$